

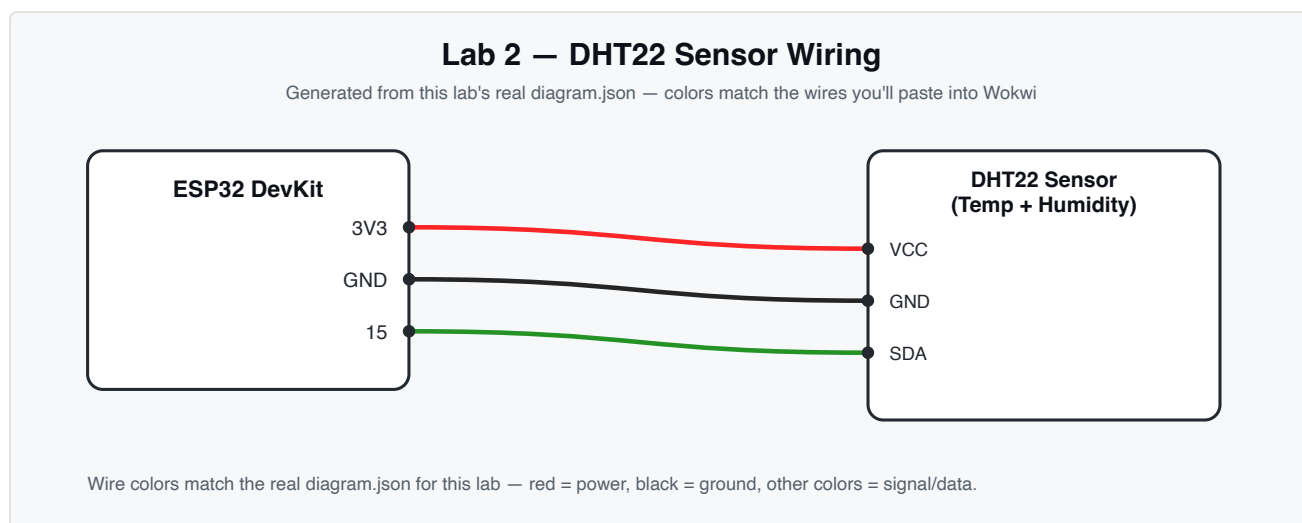
Lab 2 — Reading a Real Sensor: DHT22 (Temperature & Humidity)

Time: ~25 minutes **You'll learn:** how to read a real, commonly-used sensor and turn its raw output into numbers people can understand.

What you're building

An ESP32 wired to a **DHT22** sensor — a small, cheap sensor found in real weather stations, greenhouses, and smart thermostats. It reports both temperature and humidity from one part.

Wiring diagram

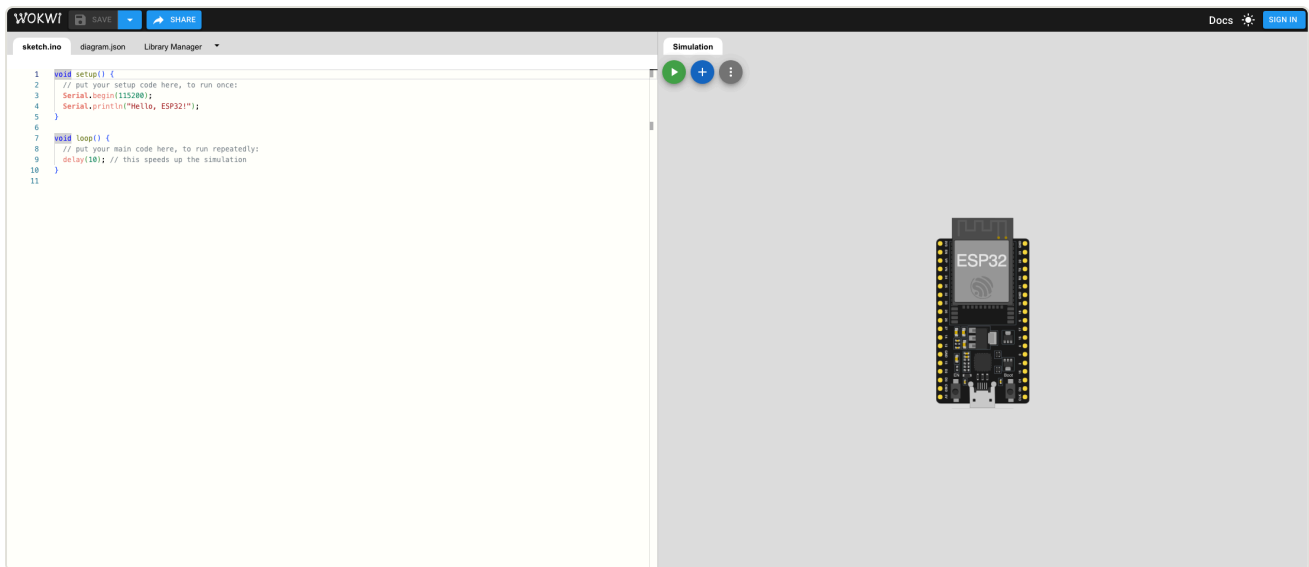


ESP32 Pin	Wire Color	Connects To
3V3	red	DHT22 Sensor (VCC)
GND	black	DHT22 Sensor (GND)
15	green	DHT22 Sensor (data pin)

Step-by-step

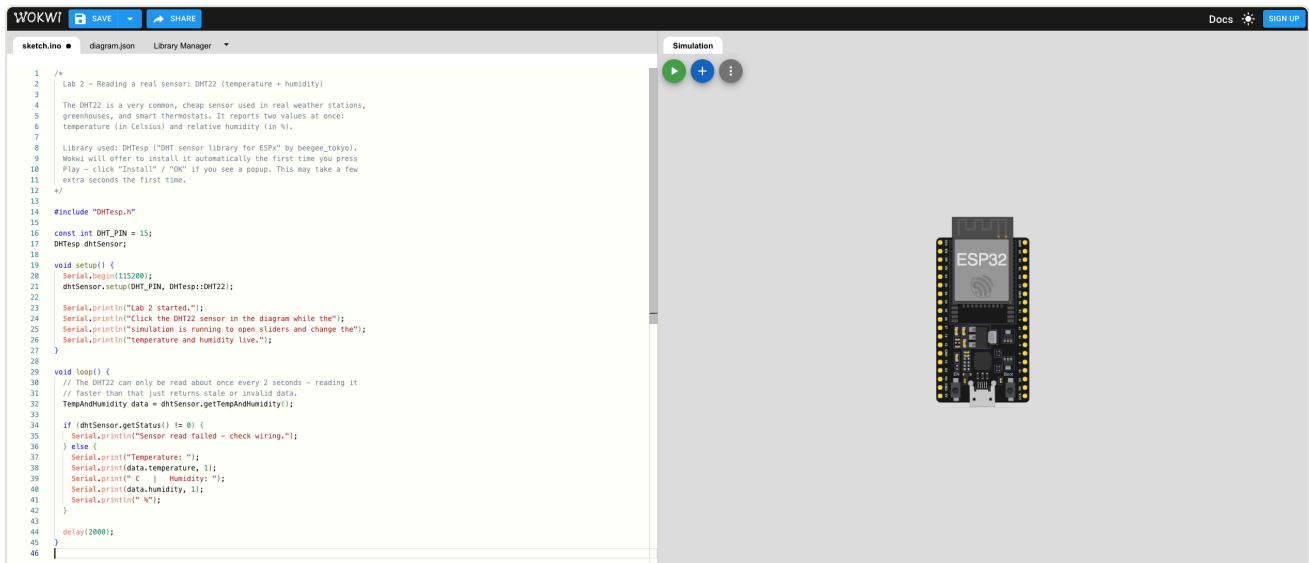
1. Open a new ESP32 project

Go to wokwi.com/projects/new/esp32 (or reuse your Lab 1 project — either is fine).



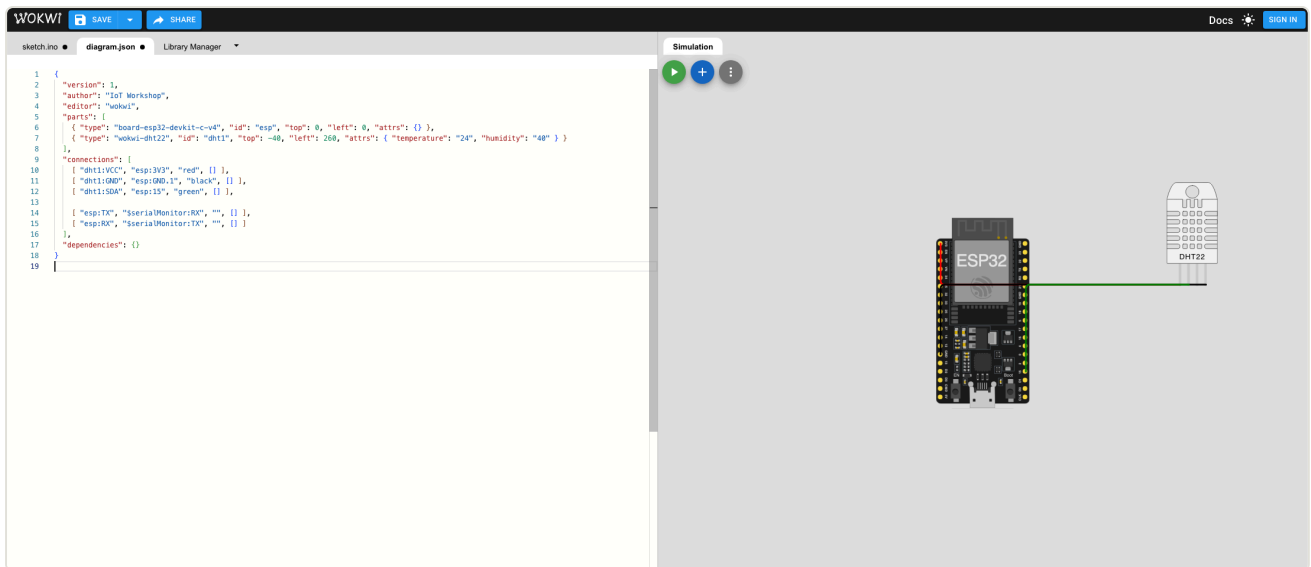
2. Paste the code

Copy `sketch.ino` from this folder into Wokwi's **sketch.ino** tab, replacing the default code.



3. Paste the circuit

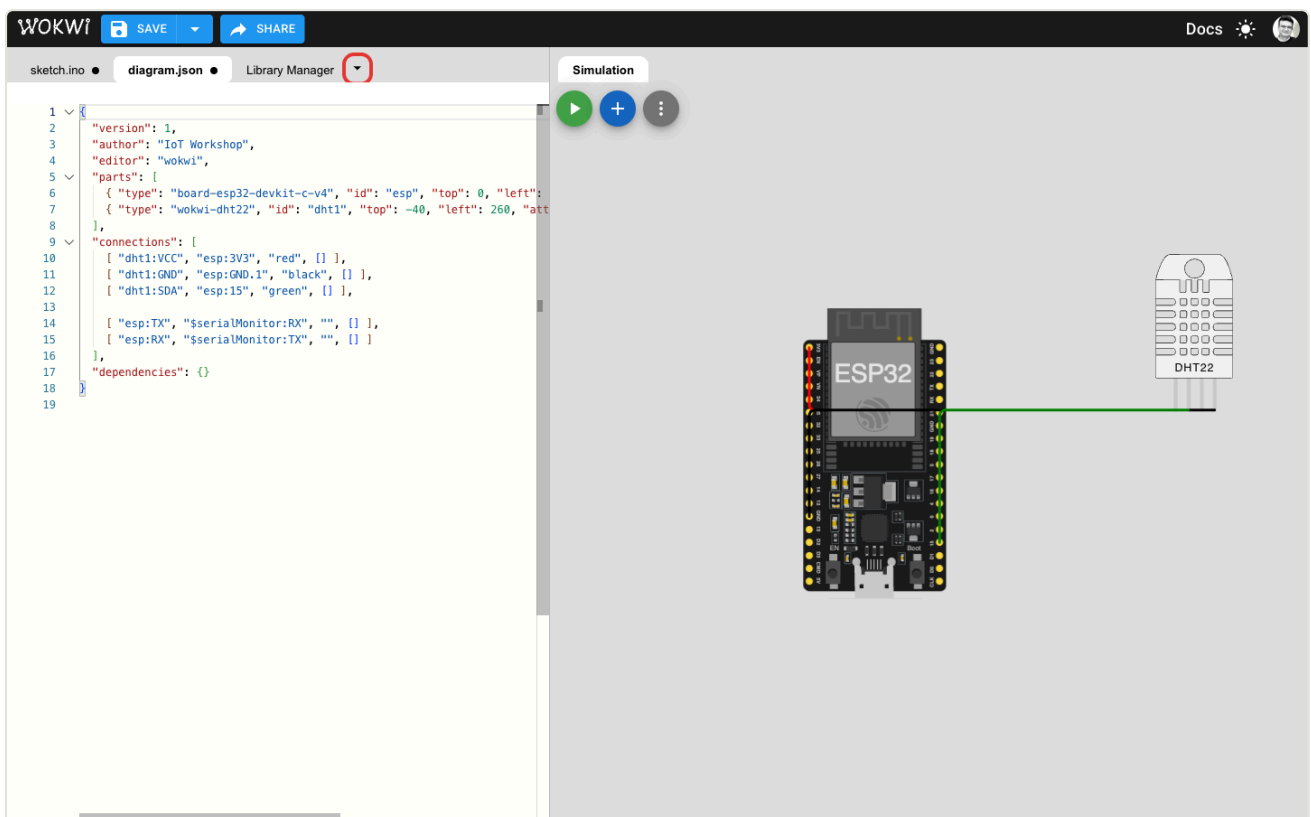
Copy `diagram.json` into Wokwi's **diagram.json** tab. The DHT22 sensor appears on the right, wired to the ESP32.



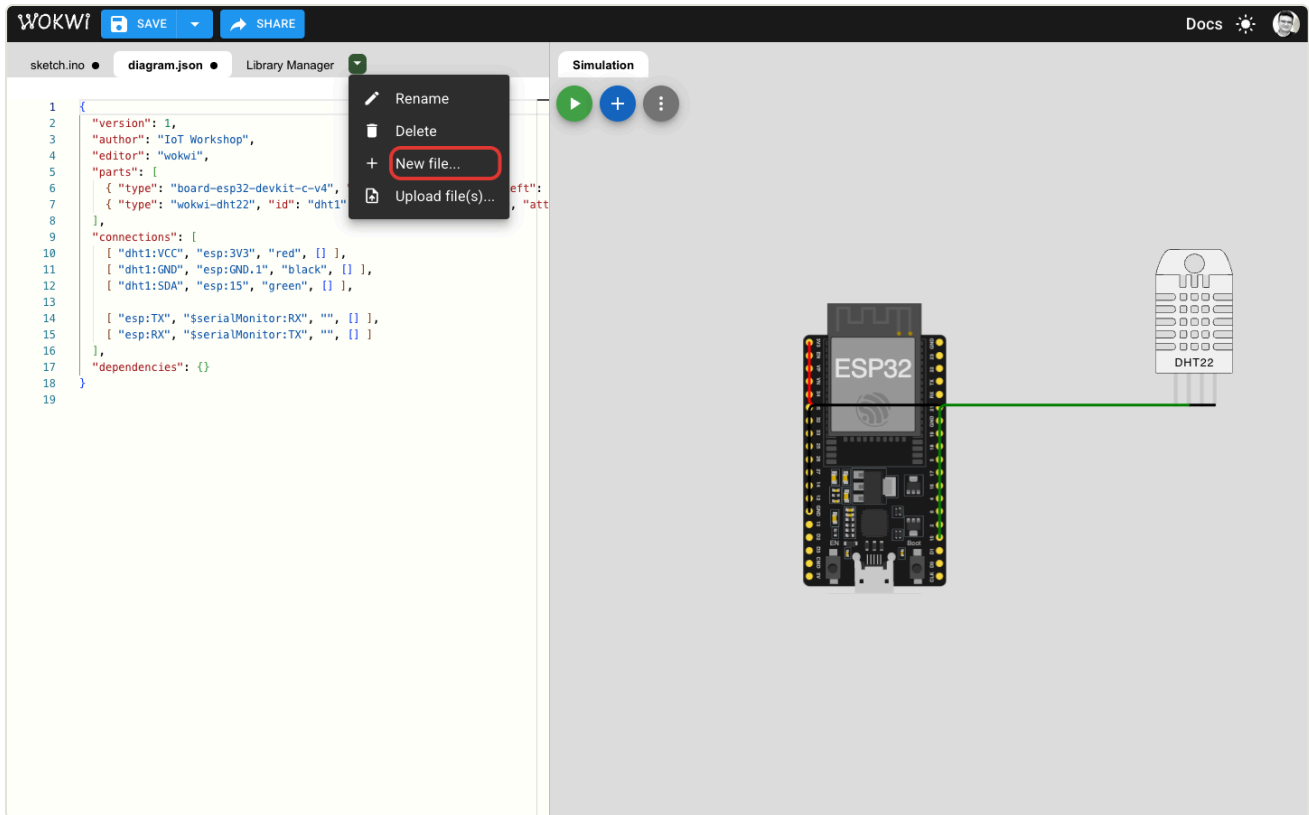
4. Add the library list

This sketch needs a library (`DHTesp`) that Wokwi does **not** install automatically just because the code includes it — you have to tell it what to install, once, via a small extra file.

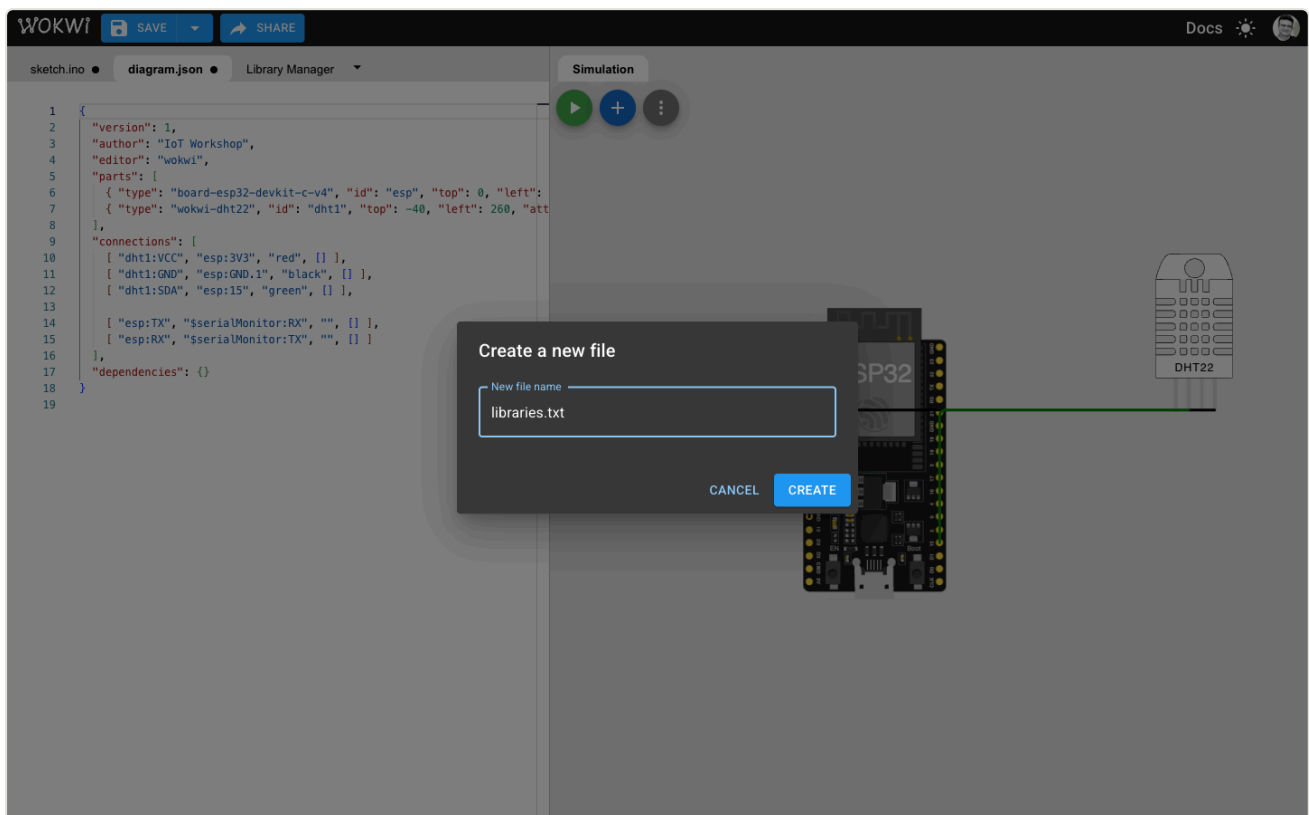
1. Click the small ▼ next to the **Library Manager** tab.



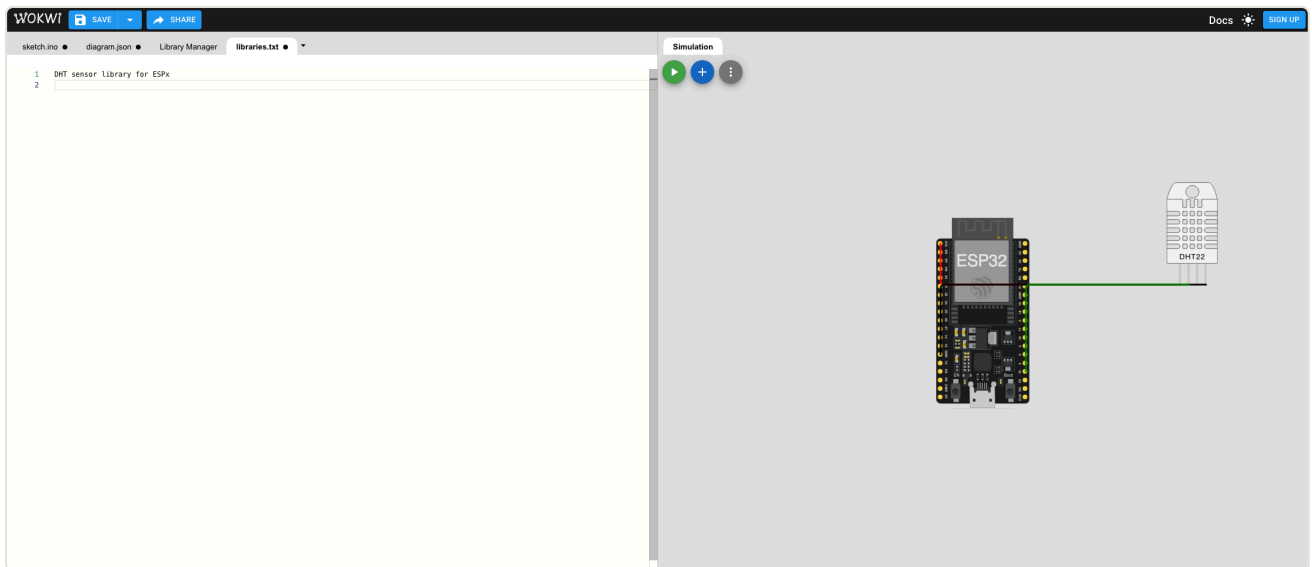
2. Choose **New file....**



3. Type the name exactly `libraries.txt` and click **CREATE** (or press Enter).

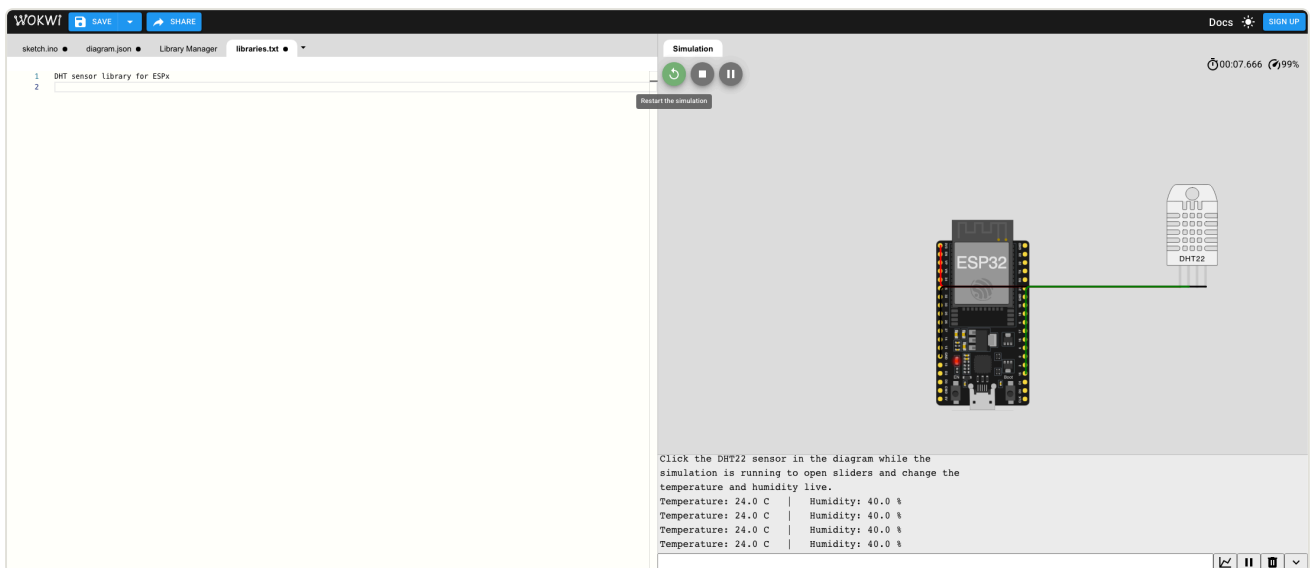


4. Copy the contents of `libraries.txt` from this folder and paste them into the new `libraries.txt` tab.



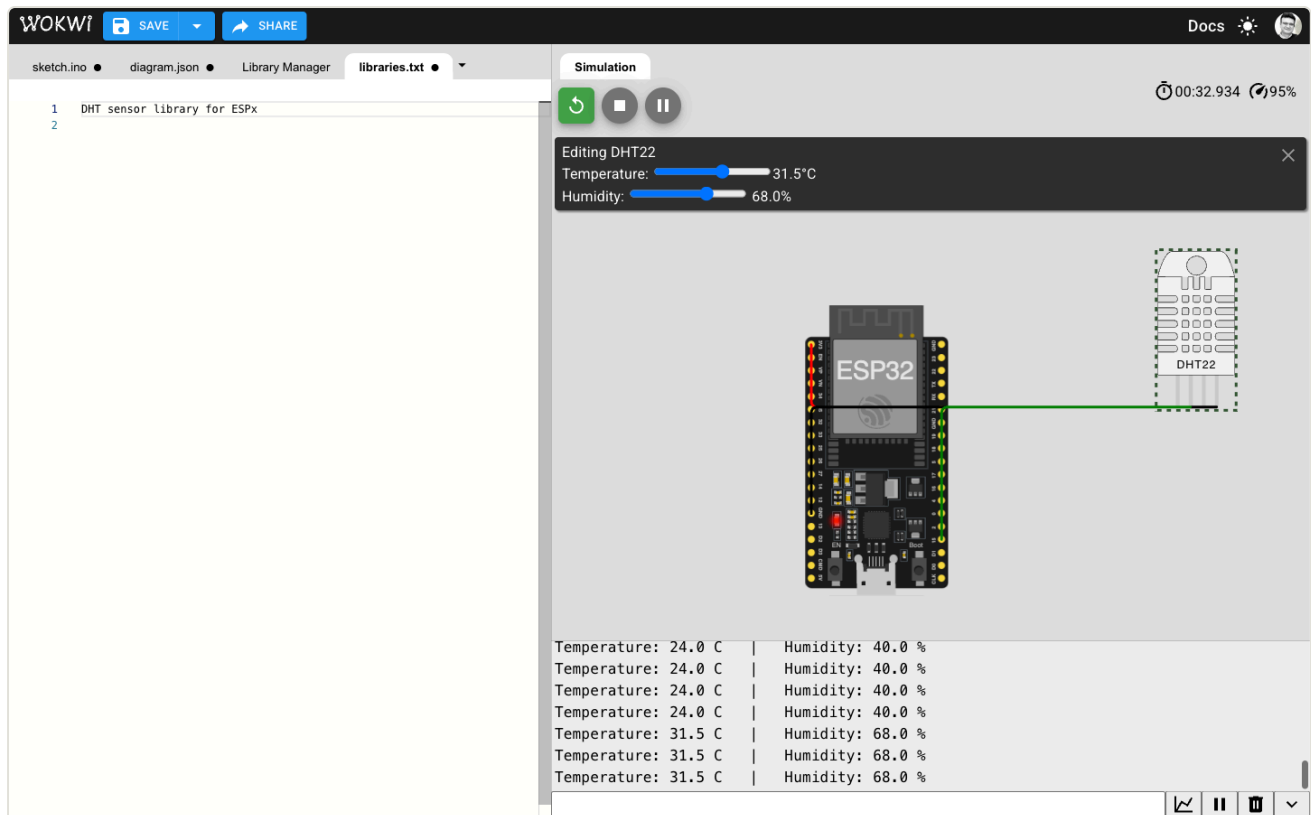
5. Press Play and watch it run

Click the green ► **Play** button. The first compile of a project with a library can take longer than Lab 1's — Wokwi has to fetch and build the library too, not just your code. This is a one-time cost per project.



Try it

- While the simulation runs, **click the DHT22 part** in the diagram. A popup with two sliders (temperature, humidity) appears. Drag them and watch the Serial Monitor readings change to match, within a couple of seconds.



- This is exactly what happens with a real sensor: the code doesn't know or care *why* the value changed (a slider here, a warm room in real life) — it just reads whatever the sensor currently reports.

What you should see

Lab 2 started.

Click the DHT22 sensor in the diagram while the simulation is running to open sliders and change the temperature and humidity live.

Temperature: 24.0 C | Humidity: 40.0 %

Temperature: 24.0 C | Humidity: 40.0 %

24.0°C / 40% is the DHT22's default simulated state — it'll stay exactly there until you drag the sliders.

Understand the code

- `dhtSensor.setup(DHT_PIN, DHTesp::DHT22)` tells the library which pin the sensor is on, and which exact sensor model it is (there's also a DHT11, a cheaper/less accurate cousin).
- `dhtSensor.getTempAndHumidity()` returns both values at once, bundled into one `TempAndHumidity` result.

- The `delay(2000)` matters: the DHT22 physically cannot be read faster than about once every 2 seconds. Reading it too often is a very common beginner bug that produces garbage readings.

Troubleshooting

- **Build failed! fatal error: DHTesp.h: No such file or directory.** You skipped step 4, or `libraries.txt` is empty/misspelled. Re-check it contains exactly `DHT sensor library for ESPx` on its own line.
- **First Play takes a couple of minutes, or shows “Build Servers Busy.”** This is normal for a library-heavy sketch on Wokwi’s free tier — it isn’t frozen. Wait it out or click Play again.
- **Sensor readings never change.** Make sure you’re dragging the sliders inside the popup (click the DHT22 part first to open it), not just hovering over the part.

Challenge (if you finish early)

- Add a rule: print `"It's cold in here!"` if temperature drops below 18°C, and `"It's warm in here!"` above 28°C. (This is a sneak preview of Module 6 — automation.)
- Convert the temperature reading to Fahrenheit as well, and print both.